AI DATA ANALYST AGENT

Professional Data Analysis Report

Generated On: 25 July 2026 04:00 PM

Developer: Pritiranjan Rout

Version: 2.0.0

EXECUTIVE SUMMARY

This business report evaluates an employee dataset of 102 records across various departments, cities, and performance metrics. The data has a health score of 86, indicating generally good quality but highlighting specific areas of concern including minor missing data and duplicate entries. The objective of this report is to outline data health, analyze workforce distribution, and provide actionable recommendations to optimize database integrity and workforce performance.

REPORT INFORMATION

Property	Value
Generated By	AI Data Analyst Agent
Developer	Pritiranjan Rout
Version	2.0.0
Generated On	25 July 2026 04:00 PM

DATASET SUMMARY

Metric	Value
Rows	102
Columns	7
Health Score	86%
Duplicate Rows	2

DATASET HEALTH SCORE

Overall Health Score : 86%

MISSING VALUES

Column	Missing
EmployeeID	0
Age	0
Department	1
Salary	1
Experience	0
Performance	0
City	0

OUTLIER ANALYSIS

Column	Outliers
EmployeeID	0
Age	0
Salary	0
Experience	0

STRONG CORRELATION ANALYSIS

Column 1	Column 2	Correlation
No Strong Correlation	-	-

DATASET OVERVIEW

The dataset contains 102 rows and 7 columns: EmployeeID, Age, Department, Salary, Experience, Performance, and City. It captures key workforce attributes including demographic indicators (Age, City), professional status (Department, Experience, Salary), and qualitative outcomes (Performance rating). A preview indicates the data represents major metropolitan hubs such as Bengaluru, Delhi, and Mumbai, spanning departments like Sales, Finance, and IT.

DATA QUALITY ANALYSIS

The dataset has a quality health score of 86 out of 100. There are 2 missing values: 1 in Department and 1 in Salary. Furthermore, 2 duplicate rows exist within the 102 records. No significant outliers were detected across key numeric fields like Age, Salary, and Experience, which suggests a relatively clean distribution of baseline records despite the duplicates and missing values.

STATISTICAL ANALYSIS

An initial review of the statistical distribution reveals zero strong correlations among the numerical attributes (Age, Salary, and Experience). While experience levels vary widely (ranging from 1 year to 34 years in the preview), they do not show a direct linear relationship with salaries or ages based on the lack of strong correlation metrics. Further distribution analysis via statistical plotting is required to understand performance and compensation bands.

BUSINESS INSIGHTS

The workforce is highly distributed across key metropolitan cities (Bengaluru, Delhi, Mumbai) and key business units (Sales, Finance, IT). Initial preview data reveals a broad span of experience (1 to 34 years) and performance levels (High, Medium, Low), suggesting a diverse workforce. The lack of strong correlation between salary and experience or age highlights that compensation structures may be driven by factors outside basic tenure, such as performance or specific department market rates.

RISK ASSESSMENT

The primary operational risks stem from data integrity issues, specifically the 2 duplicate rows and the 2 missing fields (Department and Salary). Missing salary and department data can lead to skewed average compensation calculations and incorrect departmental budgeting. Furthermore, duplicate records pose a risk of double-counting employees in performance and headcount reports, potentially leading to incorrect strategic decisions.

AI RECOMMENDATIONS

First, clean the dataset by resolving the 2 missing values through targeted database lookups or imputation. Second, identify and remove the 2 duplicate rows to prevent distorted analyses. Third, implement data entry validation rules to prevent future duplicates or missing entries at the source. Fourth, execute the recommended visual analyses (histograms, scatter plots, and count plots) to establish clear baselines for workforce metrics and compensation equity.

Rule-Based Recommendations

✓ ■■ Dataset contains 2 missing values. Consider filling or removing them.

✓ ■■ Dataset contains 2 duplicate rows. Consider removing them.

✓ ■ No significant outliers detected.

✓ ■ Histogram, Box Plot and KDE Plot are recommended for numeric columns.

✓ ■ Pie Chart, Donut Chart and Count Plot are recommended for categorical columns.

✓ ■ Scatter Plot and Bubble Chart are recommended to analyze relationships.

CONCLUSION

In conclusion, while the workforce dataset is generally healthy with a score of 86, addressing the minor data quality anomalies is critical before using this data for high-stakes HR decisions. Once duplicates are removed and missing data is resolved, the dataset will serve as a reliable foundation for deep-dive workforce analytics, helping align compensation strategies and performance management across departments.